

SDD Developer Assessment

Employee Internal Transfer Digital Journey

1. Objective

This exercise evaluates your ability to apply INT's Specification-Driven Development/Delivery (SDD) methodology to an enterprise employee digital journey.

The focus is not only on implementation.

You are expected to demonstrate how you:

- Understand a business journey.
- Identify ambiguity and missing decisions.
- Convert business requirements into a precise specification.
- Define testable acceptance criteria.
- Define API contracts.
- Design the technical approach.
- Decompose the work into independently verifiable tasks.
- Apply test-first development.
- Handle integration, security and failure scenarios.
- Maintain traceability from requirement to implementation.

The expected SDD chain is:

Business Requirement → Spec → Gate 1 → Plan → Tasks → Test First → Implementation → Gate 2 → Release

2. Business Context

The organisation has a One-Point Employee Portal through which employees can access HR, payroll, IT, learning, facilities and other employee services.

Currently, an employee requesting an internal transfer needs to interact with multiple teams and systems.

For example:

1. Employee discusses the transfer with the manager.
2. Manager confirms the transfer.
3. HR validates eligibility.
4. Employee's organisational information is updated.

5. Payroll may need to be updated.
6. IT may need to provision or remove access.
7. Facilities may need to arrange the employee's new location.
8. Employee receives confirmation.

The organisation wants to provide a single digital journey through the One-Point Employee Portal.

3. Business Requirement

An employee should be able to initiate an Internal Transfer Request from the One-Point Employee Portal.

The employee should be able to:

- Select the proposed new department/business unit.
- Select the proposed new location.
- Select the proposed role/job position.
- Provide an effective date.
- Provide an optional reason for the transfer.
- Submit the request.
- View the current status of the request.
- View actions that are pending with other stakeholders.

The organisation wants the portal to orchestrate the downstream activities and provide the employee with a single view of progress.

4. Developer Assignment

Using the INT SDD methodology, produce the following artefacts.

- Deliverable 1 — Requirement / Discovery Analysis

Identify:

- Business objective
- Primary users
- Journey stages
- Business rules
- Known decisions
- Open questions
- Assumptions
- Dependencies
- Out-of-scope items

Clearly distinguish:

Business decision vs Technical decision

- Deliverable 2 — Feature Specification in .md file and Acceptance Criteria [Define individually identifiable acceptance criteria]
- Deliverable 3 — Spec-Derived Test Cases
- Deliverable 4 — Technical Plan
- Deliverable 5 — Task Decomposition
- Deliverable 7 — AI Prompts
- Deliverable 8 — Security Assessment
- Deliverable 9 — Gate 1 Review
- Deliverable 10 — Gate 2 Evidence

5. Evaluation Criteria

Area	Weight
Business journey understanding	10%
Ambiguity & discovery	15%
Specification quality	20%
Acceptance criteria & testability	15%
Task decomposition	10%
Test-first approach	5%
Security & failure handling	5%
SDD traceability	20%
Total	100%

6. Recommended 10-day timeline

Day	Activity	Expected Output	Approx. Effort
Day 1	Understand requirement + Discovery	Business understanding, questions, assumptions, open decisions	2.5-3 hrs
Day 2	BRD interpretation + SDD Spec	BRD interpretation + initial .spec.md	3-4 hrs
Day 3	Acceptance Criteria + API Contract + Test Cases	Complete specification + AC + API + spec-derived tests	3-4 hrs

Day 4	Gate 1 Peer Review	Review comments + revised/approved spec	1.5-2 hrs
Day 5	Technical Plan + Architecture	.plan.md, integration approach, data model, failure handling, ADR candidates	3-4 hrs
Day 6	Task Decomposition + AI Prompts	.tasks.md, task-to-AC mapping, implementation prompts	2.5-3 hrs
Day 7	Test-first implementation	RED tests + first implementation increment	4-5 hrs
Day 8	Guided implementation	Remaining core implementation + integration	4-5 hrs
Day 9	Validation + Security + Gate 2 preparation	GREEN tests, security review, traceability	4-5 hrs
Day 10	Gate 2 + Final Presentation	Final artefact chain + demo + 10-15 min walkthrough	2.5-3 hrs

7. Milestones

Assessment is segregated into 4 major milestones:

Milestone 1 — Discovery & Specification: Days 1-2

The developer should demonstrate that they don't start coding immediately.

By the end of **Day 3**, they should have:

BRD → Spec → AC → API Contract → Test Cases

Milestone 2 — Gate 1: Day 3

This should be a real peer review, not merely a presentation.

Milestone 3 — Plan → Tasks: Days 4-5

Milestone 4 — Implementation & Gate 2: Days 6-8

We would require:

Task → Prompt → Test (RED) → Implementation → Test (GREEN) → Review

Effort expectation of approximately 30-35 hours total, spread over 8 working days